

Methodology: Smart Product Pricing Challenge

1. High-Level Strategy: State-of-the-Art Multi-Modal Ensemble:

This solution is a hybrid model designed for a top-tier score by leveraging fine-tuned deep learning models for feature extraction, followed by a powerful gradient boosting ensemble. The architecture is built on the principle of Feature Fusion, where expert models for language and vision are custom-trained on the competition data to create rich, high-dimensional embeddings.

2. Feature Engineering & Preprocessing:

Three distinct, SOTA feature sets were generated as the foundation for the final model.

SOTA Vision Features (Fine-Tuned CNN):

- **Model:** A powerful **EfficientNetB3** was used as the vision expert.
- **Fine-Tuning:** The deeper layers of the pre-trained model were unfrozen and **fine-tuned** on the 75,000 product images with data augmentation. This specialized the model to recognize subtle, price-relevant visual cues like packaging quality and material texture.
- **Output:** A **1536-dimension embedding vector** was extracted from the model for each product image.

SOTA Textual Features (Fine-Tuned LLM):

- **Model:** The microsoft/deberta-v3-base Transformer, a state-of-the-art Large Language Model, was used as the language expert.
- **Fine-Tuning:** The model was fine-tuned for regression on the 75,000 catalog_content descriptions, teaching it the deep, contextual meaning of e-commerce language.
- **Output:** A 768-dimension embedding vector was extracted from the [CLS] token's final hidden state for each product description.

3. Final Predictive Model: A Blended Ensemble:

- **Algorithms:** Two world-class gradient boosting models were trained on the final, fused feature set (Vision Embeddings + Language Embeddings + Numerical Features):
 1. **LightGBM Regressor**
 2. **XGBoost Regressor** (using `tree_method='gpu_hist'` for GPU acceleration)
- **Ensembling Strategy:** The predictions from both models were blended using an optimized weighted average. The optimal weight was determined on a validation set to find the mixture that produced the lowest SMAPE score, resulting in a final model more accurate than either of its components.